The Kynurenine connection: how exercise shifts muscle tryptophan metabolism and affects energy homeostasis, the immune system, and the brain.

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Abstract

Tryptophan catabolism through the kynurenine pathway generates a variety of bioactive metabolites. Physical exercise can modulate kynurenine pathway metabolism in skeletal muscle, and thus change the concentrations of select compounds in peripheral tissues and in the central nervous system. Here we review recent advances in our understanding of how exercise alters tryptophan-kynurenine metabolism in muscle and its subsequent local and distal effects. We propose that the effects of kynurenine pathway metabolites on skeletal muscle, adipose tissue, immune system, and the brain, suggest that some of these compounds could qualify as exercise-induced myokines. Indeed, some of the more recently discovered biological activities for kynurenines include many of the best-known benefits of exercise: improved energy homeostasis, promotion of an anti-inflammatory environment, and neuroprotection. Finally, by considering the tissue expression of the different membrane and cytosolic receptors for kynurenines, we discuss known and potential biological activities for these tryptophan metabolites.

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INTRODUCTION

Skeletal muscle, one of the largest organs and the engine of the locomotor system in our body, accounts for up to 40% body mass in a lean individual (66). Together with the liver, our muscles take up most of the post-prandial glucose in response to insulin stimuli (115) and significantly contribute to maintaining body temperature (7). Additionally, skeletal muscles constitute a large reservoir of amino acids that in critical conditions can be displaced for energy production by various organs (10). Skeletal muscle is a highly plastic organ, whose composition and performance can be modified through physical exercise. The benefits of exercise go beyond muscle function and systemic energy homeostasis and include, for example, improvements in brain plasticity and cognition (61). For that reason, physical exercise can be used to prevent and treat a variety of diseases such as type 2 diabetes (32, 150), breast and colon cancer (150, 155), depression (133), cancer cachexia (44), and sarcopenia (107).

As we try to understand the precise mechanisms that underlie the many effects of exercise, it is important to note that although both physical activity and physical exercise are beneficial to health, they are not the same thing and can involve very different molecular pathways. Physical activity is defined as any movement that involves skeletal muscles and results in energy expenditure, including daily life activities. Exercise, however, refers to planned, structured, and often repetitive movements that aim to improve or maintain a certain level of performance (15). Exercise has type-specific benefits for human health and can be used to ameliorate different conditions. Endurance training increases muscle energy efficiency, oxidative capacity, resistance to fatigue as well as cardiovascular function, whereas resistance training leads to muscle hypertrophy and can be used to treat sarcopenia (for review see (29, 53). Finally, when describing exercise, we will use acute to describe a single bout of exercise and chronic/training to describe a prolonged exercise routine.

In addition to its importance for locomotion and energy utilization, skeletal muscle is increasingly appreciated as an endocrine organ with the capacity to secrete a variety of factors (myokines). These metabolites and proteins exert autocrine, paracrine, or endocrine effects and affect metabolism, inflammation, cell proliferation, among other processes (72).

This review's purview will be the effects of exercise and the gene regulatory and biochemical pathways that change between the untrained and trained state, focusing on the exercise-induced modulation of the tryptophan-kynurenine metabolism and its local and systemic effect. Here we capture the current state of this relatively young research field, but many unknowns remain.

REGULATORS OF SKELETAL MUSCLE ADAPTATION TO EXERCISE

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In order to cope with repeated exercise challenges, muscle fibers undergo structural and metabolic adaptations to ensure the right fuel is supplied in the correct amount to the appropriate contractile apparatus. Skeletal muscle is composed of different types of fibers, whose nomenclature is classified based on their metabolic structure and type of myosin heavy chain expressed. The two extremes of fiber types are represented by the glycolytic, high force developing, fast-twitch, fatigable type IIB fibers, versus the oxidative, fatigue resistant, slow-twitch, low force developing type I fibers (126). These fibers are optimized to utilize different fuel sources. Type IIB fibers have higher levels of glycogen storage and can generate more ATP from glycolysis (compared to type I) due to enhanced lactate dehydrogenase activity, elevated glycerol-3-phosphate shuttle activity, and increased levels of phosphofructokinase/fructose-bisphophatase 3 (126). Type I fibers, on the other hand, have higher lipid stores and elevated expression of lipases to utilize fatty acids for oxidative phosphorylation.

Different exercise training programs can be used to improve particular aspects of muscle function. These training-induced, long-lasting adaptations involve changes in the expression of gene networks coordinated by transcription factors and coregulators. Transcription factors respond to specific stimuli and can directly regulate the expression of genes involved in the adaptation to that stimulus. However, for the muscle fiber to functionally adapt to a new level of performance, it requires a high level of coordination between innervation, vascularization, energy metabolism, calcium handling, and the expression of the right contractile apparatus. For this reason, coactivator-mediated regulation of gene expression has emerged as an important mechanism in muscle adaptation to exercise, by ensuring that the activity of the different transcription factors is enhanced in a coordinated way.

Peroxisome Proliferator-Activated Receptor γ coactivators (PGCs)

The PGC family of transcriptional coactivators is important for skeletal muscle function and adaptation to chronic exercise. The family includes PGC-1α, PGC-1β, and PGC-related Coactivator (PRC) (20, 39, 83), which enhance the activity of nuclear hormone receptors and other transcription factors. PGC-1α is the most studied member and is highly regulated in a context- and tissue-specific manner. For example, skeletal muscle PGC-1α expression is induced by acute and chronic exercise while in the liver it responds to fasting. Interestingly, the PGC-1α gene is expressed as a collection of splicing variants encoding proteins with different biological activities (for review (83). Our laboratory has focused on understanding the regulation and activity of four of these PGC-1α variants (PGC-1α1 through α4; (121). PGC-1α1 represents the canonical and longest transcript variant, initially identified as a cold-inducible gene expressed in brown adipose tissue (110). PGC-1α2 and α3 are involved in the regulation of alternative splicing (82), and PGC-1α4 is the shortest known transcript of the family, primarily activated during resistance exercise, providing adaptations that lead

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to muscle mass and strength increase. The hypertrophic phenotype is mainly achieved through the induction of IGF1 and repression of myostatin expression (121).

In skeletal muscle, PGC-1α1 activation drives a genetic program that promotes fiber type-switching, angiogenesis, and mitochondrial biogenesis, providing the structural and functional adaptation to endurance exercise (18, 106, 110, 127). PGC-1α1 activity is also involved in the regulation of numerous metabolic pathways, such as branched-chain amino acid metabolism, alanine metabolism, pentose phosphate pathway, fatty acid oxidation, TCA cycle and others (50–52). Importantly, in addition to regulating local adaptations in the muscle fibers where they are induced, PGC-1a coactivators also control the expression and release of several myokines such as Irisin (12), β-Aminoisobutyric acid (BAIBA), meteorin-like (113), neurturin (91), and IL-6 (48). The release of these mediators is higher after acute exercise and believed to play a major role in muscle to other organ cross-talk (101, 116, 134, 156).

THE KYNURENINE PATHWAY OF TRYPTOPHAN METABOLISM

Tryptophan (Trp) is an essential amino acid required for protein synthesis and has two prominent degradation pathways. The first is towards serotonin biosynthesis and its derivative melatonin, which are involved in the regulation of mood and sleep, respectively (22, 31). The second is the kynurenine pathway, which can lead to the formation of nicotinamide adenine dinucleotide (NAD^+) (16). The kynurenine pathway includes several enzymes that degrade 95% of the free Trp into diverse bioactive kynurenine pathway metabolites (KPMs), often referred to as kynurenines.

Kynurenine (Kyn) is the first of many bioactive metabolites along the kynurenine pathway, which have been shown to be involved in modulation of the immune and central nervous systems (138, 149). For this reason, Kyn and KPMs have been mostly studied in the context of cancer and psychiatric disease (14, 93). The conversion of Trp to Kyn requires the enzymes tryptophan 2,3-dioxygenase (TDO) or indoleamine 2,3-dioxygenase (IDO) and kynurenine formamidase and occurs mainly in the liver. However, under high cortisol and inflammatory cytokine levels, hepatic TDO expression is repressed and immune cell IDO in induced. This shifts Kyn production to the immune compartment as part of a negative feedback loop in place to control inflammatory responses. In skeletal muscle there is little expression of IDO (**Table 1**) and minimal TDO enzymatic activity (103), so Kyn is more likely produced in the liver (or immune cells) and then imported into muscle fibers. Kyn can be then metabolized into kynurenic acid (KynA), Anthranilic Acid (AA) or 3-hydroxykynurenine (3-HK). Interestingly, the PGC-1α1/PPARα/δ axis in muscle increases the expression of kynurenine aminotransferases (KATs), which convert Kyn to KynA. These enzymes belong to the kynurenine pathway of tryptophan degradation and were shown to be increased in skeletal muscle after exercise in both mice and humans (1).

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Some of the kynurenine pathway compounds are end-metabolites and some can be further processed to NAD^+ (Figure 1). However, with the exception of hepatocytes, there are few cells in the body fully equipped with the enzymatic machinery to fully degrade Trp to NAD^+. This makes KPMs excellent candidates for mediators of inter-cell or inter-organ crosstalk, as they can be exchanged between tissues to either continue their voyage down the pathway or exert a biological effect. The biological activity of many of these metabolites is well established in the central nervous system (136, 149), while less is known in the periphery.

Import of tryptophan metabolites into skeletal muscle

In muscle fibers, Kyn and Trp uptake occurs via large neutral amino acid transporters (LATs) in open competition with other amino acids (leucine, isoleucine, methionine, phenylalanine, and tyrosine) (129) (Figure 1). Accordingly, systemic inhibition of LATs has been shown to reduce intramuscular Kyn and KynA concentrations (even under supra-physiological concentrations of Kyn) (128). In skeletal muscle, LAT1 (SLC7A5) and LAT2 (SLC7A8) form heterodimer complexes with CD98 (SLC3A2) (40), with type 2 fibers expressing more LAT1 than type 1 fibers (58). LAT1 and CD98 expression has been shown to increase in response to acute resistance exercise in young and older adults (26). Additionally, numerous studies included in MetaMEx (108) show increases in LAT1 and/or CD98 mRNA after acute exercise (resistance and endurance) (GSE 33603, 41769, 27285, 43856, 71972, 19062, 23697, and 28422).

There are a few reports of transporters that could import other KPMs into cells. LATs have been linked to 3-HK uptake in striatal neurons (98) and organic anion transporters 1 and 3 (OAT1/SLC22A6 and OAT3/SLC22A8) can transport KynA and xanthurenic acid (XA) (144, 145). Both OAT1 and OAT2 have been shown to be present in human muscle fibers by immunohistochemistry (139). However, given 1) the low plasma concentrations of KynA and XA (nM range vs µM for both Kyn and Trp) and 2) the LATs competitive inhibition by amino acids (129), the vast majority of the KPMs are likely synthesized intramuscularly (**Figure 1**).

Intramuscular kynurenine pathway enzymes

Within the muscle fiber the most abundant enzymes of the kynurenine pathway are KATs, kynureninase (KynU), kynurenine 3-monooxygenase (KMO) and 3-hydroxyanthranilate 3,4-dioxygenase (3-HAO) (**Table 1**). There are four described KATs; KAT I (also known as glutamine transaminase K and cysteine conjugate-beta lyase (CCBL) 1), KAT II (aminoadipate aminotransferase); KAT III (CCBL2); and KAT IV (glutamic-oxaloacetic transaminase (GOT) 2 and mitochondrial aspartate aminotransferase (mAST)). The previous litany of appellations should give an indication of how promiscuous these enzymes are. In depth analysis of the KATs will not be discussed here (for review see: (47, 118)). For the purpose of this review, it is relevant to know that KATs utilize a cofactor, pyridoxal-5'-phosphate (PLP, the active form

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of Vitamin B_6), to transfer an amino group from one substrate to another. In the case of Kyn, the loss of an amino group forms an intermediate molecule that undergoes intramolecular cyclization to form KynA (47). While other amino acid – α-ketoacid pairs utilized by KATs are reversible, Kyn to KynA is a terminal reaction. Additionally, many of the KATs can also utilize 3-HK as an amino donor (123), forming XA in the process. In rodent skeletal muscle, both KynA and XA are present in resting muscle at higher concentrations than their precursors Kyn and 3-HK (103), suggesting that KATs are utilizing Kyn and 3-HK at rest. Since KAT expression in muscle is highly dependent on PGC-1α1 (1), this likely originates from oxidative fibers which have naturally high levels of the coactivator.

KMO is a FAD and NADPH-dependent enzyme localized on the mitochondrial outer membrane that catalyzes the hydroxylation of Kyn to 3-HK in presence of oxygen (5, 42, 97). KMO activity has been linked with neurodegenerative diseases and acute pancreatitis (5, 63). KMO is located at the critical branching point of the kynurenine pathway that determines the fate of Kyn to neurotoxic compounds (3-HK and quinolinic acid (QA)) or to the beneficial KynA. Indeed, pharmacological inhibition of KMO activity has been shown to dramatically increase levels of KynA, a key target for treating certain neurological diseases (5). KynU is a PLP-dependent enzyme that catalyzes the cleavage of 3-HK into 3-hydroxyanthranilic acid (3-HAA) as well as Kyn into AA (4). However, the human enzyme shows selectivity towards the substrate 3-HK (151) but we cannot exclude differences between species. Both KynU-mediated reactions lead to the release of the amino acid L-alanine. As an aside, skeletal muscle alanine production is initially increased during exercise, and alanine is an important amino acid in skeletal muscle – liver crosstalk during situations of energy crisis (such as fasting) (84). However, there is little information on if KynU produced alanine is important/critical for muscle function.

KynU activity is inhibited by estrogens and is particularly sensitive to PLP levels (8). Subjects that have an inactive form of the enzyme have increased plasma concentrations of 3-HK (161-fold higher) and lower amounts of the end product NAD(H) ($1/7^{th}$ of the normal levels) (131). 3-HAO is a non-heme ferrous-ion (Fe^{2+})-containing enzyme that catalyzes, in presence of oxygen, the oxidative ring opening of 3-HAA to an unstable derivative, 2-amino-3-carboxymuconate semialdehyde, which spontaneously cyclizes to QA (8). Similarly, patients carrying a defective form of the enzyme present higher level of the substrate 3-HAA (64 to 385 times) and a lowered amount of NAD(H) ($1/4^{th}$ normal levels).

EFFECTS OF EXERCISE ON INTRAMUSCULAR KYNURENINE PATHWAY METABOLITES

Re-routing Trp metabolites could be one of the mechanisms by which our body benefits from exercise, yet how the metabolite concentrations change due to exercise is poorly understood. What is consistently reported in literature is that KynA is elevated in circulation following acute endurance exercise (1, 127). Additionally, chronic endurance exercise in humans increases the expression of all four KATs (1, 127), with the effects of resistance

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exercise on KAT expression presently unclear. In mice, chronic endurance exercise, or PGC-1α1 overexpression in skeletal muscle, enhances KAT 1, 3, 4 and 3-HAO expression, with no change to KynU or KMO (1, 2). Furthermore, muscle-specific genetic deletion of PGC-1α in mice inhibits both exercise-induced expression of KATs and elevated plasma KynA (2). At the moment, it is not known how relative amounts of Kyn, KynA and other Trp-metabolites vary in the skeletal muscle fibers following exercise.

Physical exercise could modulate the output of the kynurenine pathway even without regulating the expression of genes encoding its enzymes. KMO and 3-HAO require molecular oxygen in order to exert their catalytic activity, while KATs require α-ketoacids and, together with KynU, PLP. Exercise leads to increased O_2 consumption to sustain a higher energy production from the mitochondria (29). This could affect the kynurenine pathway in at least two ways: 1. The enzymatic activity of KMO and 3-HAO could be substantially reduced; 2. Reduced O_2 availability (hypoxia) inhibits a class of prolyl-hydroxylases (PHDs) that utilize α-ketoglutarate as a substrate to hydroxylate the hypoxia-inducible factor-1α (HIF-1α). At low O_2 levels PHDs are inhibited, so accumulated α-ketoglutarate could then be used by other enzymes, such as KATs. Indeed, a similar mechanism has been proposed (99), where the surplus of muscle-derived α-ketoglutarate is used by liver KATs to produce KynA, which in turns promotes ischemic preconditioning of the heart. Taken together, these events would certainly promote Kyn down the KynA and XA degradation branch over the NAD^+ route. Experiments with rats suggest that PLP is re-directed to skeletal muscle when the animals were challenged with sub-maximal exercise (46). This could fuel enzymes such as KATs and KynU that require this cofactor.

Kynurenine and skeletal muscle bioenergetics

The observation that endurance exercise training elevates the KATs in skeletal muscle and that circulating KynA increases after a single bout of exercise begs the question: How is skeletal muscle benefiting from Kyn utilization? It will take extensive carbon and nitrogen tracing studies to determine the contribution of Kyn (and 3-HK) to the myriad of KAT mediated pathways, but some studies have started to elucidate this intricate metabolic network. Recent findings from our lab show that Kyn synergizes with PGC-1α1 in regulating Malate-Aspartate Shuttle (MAS) enzymes, transporters, and metabolites. In short, the MAS utilizes the reversible reactions of malate to oxaloacetate to aspartate to 'shuttle' cytosolic NADH generated during glycolysis into the mitochondria. With the help of glutamate and α-ketoglutarate, the MAS increases glycolysis-related ATP production by transferring reducing equivalents into complex I of the mitochondrial electron transport chain (71). Interestingly, PGC-1α overexpression in muscle increases the transporters (Aspartate Glutamate Carriers SLC25A13 or SLC25A12 and 2-Oxoglutarate/Malate Carrier SLC25A11) as well as GOT1, KAT IV, and malate dehydrogenases (MDH) 1 and 2 (51), or every enzyme and transporter involved in each step of the MAS. Others have shown endurance exercise increases MAS

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components in human skeletal muscles: mRNA of GOT1, SLC25A12, SLC25A13, and KAT IV (GSE44818 (120)), protein levels of MDH2 and KAT IV (GOT2) (29), and enzymatic activity of MDH2 (mMDH), GOT1, and KAT IV (mASAT) (124, 125).

When mice with ectopic expression of PGC-1α1 in muscle (75) received a dose of Kyn (i.p.), there was a further increase in all MAS genes (except SLC25A12 and 13) (2). Importantly, on a wild-type background, Kyn alone was not sufficient to elicit changes in any MAS-associated genes. PGC-1α1 overexpression increased muscle malate, aspartate, and glutamate levels (51), which were further elevated upon Kyn administration (2). Additionally, malate and aspartate are both elevated following acute endurance exercise in rodents and humans (2, 102). In cultured myotubes, ATP production from glycolysis is improved by PGC-1α1 and further increased by Kyn supplementation (2). Conversely, treating myotubes lacking PGC-1α with Kyn resulted in decreased respiration. Finally, inhibition of the KATs blunted both myotube cellular respiration as well as malate and aspartate levels.

Kyn levels in resting skeletal muscle are much lower than glutamate and aspartate (in rodents, pmol/g (103, 128); in humans, μmol/g (11, 41, 45)), yet how Kyn levels change following exercise is unknown. It has been shown that a bolus administration of Kyn orally can cause rodent skeletal muscle concentrations to increase over 10-fold (128) and, as mentioned earlier, LATs increase following exercise. So it is possible that Kyn influx is increased in muscles during and following exercise. However, the location of Kyn (and subsequent metabolism) is just as important as concertation. For example, if all muscle fiber Kyn is converted to KynA in the mitochondria, then that could influence any mitochondrial metabolic network. Additionally, since Kyn to KynA conversion is unidirectional, there is a permanent transfer of amino groups to other metabolic pathways. Aside from oxaloacetate and α-ketoglutarate (the aspartate and glutamate α-keto acids), KATs have been shown to efficiently utilize 11 other α-keto acids (47). Tracing studies would confirm their relative importance in Kyn and 3-HK aminotransfer activity.

In addition to enhanced MAS activity and ATP production, the shifting of KPMs could have antioxidant benefits. Both KynA and XA are versatile free-radical scavengers (105), with KynA (10 μM) being able to reduce lipid oxidation (79). Not much has been done to investigate KynA and XA antioxidant effects in muscle, but KynA reduces QA-induced oxidation and mitochondrial dysfunction in rat striatal slices (38). Additionally, Kyn treatment (μM range) increased mouse and human myotube reactive oxygen species (ROS) generation and lipid peroxidation (70), and there is evidence that 3-HK can also generate ROS (98). While these effects have mostly been seen at supraphysiological levels, they could be relevant during exercise. KATs can transform ROS generators into scavengers. Since KAT IV is the most increased by exercise, KynA and XA may become concentrated inside the mitochondria, where they could act as free radical scavengers (105).

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KYNURENINE PATHWAY METABOLITES AND INTER-ORGAN COMMUNICATION FOLLOWING EXERCISE.

The link between skeletal muscle, exercise, and KPMs came from the observation that exercise trained muscle (or muscle with high PGC-1α1 levels) has very high levels of KAT enzymes (1, 127). This enhances peripheral Kyn to KynA conversion and prevents the accumulation of Kyn in the brain, which is observed in certain mental health disorders such as stress-induced depression. In mice, this detoxification mechanism was shown to reduce Kyn-related excitotoxicity, neuroinflammation, and depressive-like behaviors (1). Since the activation of this pathway is extremely dependent on PGC-1α1, mice genetically engineered to lack expression of this coactivator in muscle show depressive-like behavior and a reduced capacity to catabolize Kyn, even in the absence of any challenge. This highlighted skeletal muscle and exercise as novel regulators of Trp-Kyn metabolism, with important systemic consequences. While we know that plasma KynA (but not Kyn) increases following exercise, we still do not know how KynA leaves muscle fibers (Figure 1) or if any other KPMs are elevated following exercise (either intramuscularly or in circulation).

From muscle to adipose tissue and immune system

Since the elevation in circulating KynA following exercise is driven by skeletal muscle (1), subsequent studies aimed at investigating if KynA can work as a myokine. Indeed, the daily elevation of KynA to post-exercise levels in mice over a period of 1 to 4 weeks resulted in weight-loss due to reduction in adiposity (3). Interestingly, weight loss was not due to changes in food intake, but to an increase in adipose tissue energy expenditure mediated by induction of beige adipocytes in the subcutaneous and visceral adipose depots (a phenomenon known as adipose tissue browning). Beige adipocytes are able to "waste" energy by linking fatty acid oxidation to futile cycles that dissipate energy in the form of heat (19). KynA administration to mice also changed the immune cell signature within the adipose tissue, which shifted to a more anti-inflammatory phenotype (3). The observed effects of KynA were mediated by GPR35 signaling (which will be discussed in more detail below), and subsequent upregulation of PGC-1α1. Importantly, GPR35 knockout mice are resistant to KynA- or exercise-induced browning of the adipose tissue. This further supports the role of KynA as an exercise-induced myokine with effects on systemic energy expenditure and immune system control. In addition to GPR35 there are other KPM receptors expressed in different peripheral tissues. This suggests that other KPMs could be myokines as well (**Figure 2**).

Peripheral NMDA receptors

The *N*-methyl-D-aspartate receptor (NMDAR) is a ligand and voltage gated channel. It requires both glutamate and membrane depolarization to become a non-specific cation

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channel, though it mainly functions as a calcium channel (49). The receptor is a heterotetramer commonly comprised of two GluN1 subunits and two other identical subunits; GluN2A or B are the most frequent, with GluN2C, D and GluN3A, B also being possible (for distribution of subunits in non-neuronal tissue: (59)). Additionally, the subunit composition of the receptor influences the function (gating mechanics, glutamate sensitivity, deactivation time course, ion permeability) (49). NMDARs plays a vital role in neuron function and excitotoxicity (87). Since KynA cannot cross the blood-brain barrier (BBB), we will explore the potential effects of muscle-derived KynA as an NMDAR antagonist in the periphery.

In the pancreas, β-cell NMDARs can modulate glucose-stimulated insulin secretion and glucose tolerance (81). Specifically, genetic ablation of the β-cell GluN1 subunit in mice results in enhanced glucose-stimulated insulin secretion. Glucose-stimulated insulin secretion was also elevated in mouse and human islets treated with the NMDAR antagonists MK-801 and dextromethorphan. Further studies have shown elevated glutamate and prolonged NMDAR activation play a role in β-cell glucotoxicity, or loss-of-function due to chronic hyperglycemia (62). Again, NMDAR antagonists were able to restore β-cell function and decrease β-cell oxidative stress. It is well established that exercise is an efficient treatment for type-2 diabetes (132). Muscle-derived KynA antagonizing β-cell NMDARs could be one of the many mechanisms contributing to this benefit.

An active field of research revolves around NMDARs in the kidney, pituitary gland, and bone. A recent review showcased the potential of glutamate and NMDARs in calcium and phosphorus regulation (13). While muscle-derived KynA could play a role in mineral metabolism across these organs, we will focus on bone NMDAR. Blocking the NMDAR in primary osteoblastic cells lead to a decrease in bone resorption *in vitro* (17), a result that has been confirmed in other studies (65, 88), potentially through disruption of sealing zone formation (65) or regulation of osteoclastogenesis (104). In an animal model of osteoporosis, the drug ebselen was shown to alleviate LPS-induced bone loss and anti-osteoclastic effects by blocking NMDARs (6). On the other hand, blocking NMDAR in osteoblasts decreased matrix mineralization *in vitro* (55, 57). NMDAR inhibition via MK-801 decreased osteoblast maturation (57), and subsequent studies have shown α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid (AMPA) and NMDA increases ERK phosphorylation and osteocalcin mRNA in osteoblasts (76). Again, blocking either the AMPA or NMDA receptors inhibited calcium deposition and alkaline phosphatase activity. As an interesting side note, KynA has been shown to have a dual effect on the AMPAR, potentiating the receptor current in low (μM and below, values expected post exercise) and antagonizing the receptor at high, non-physiological doses (mM) (109). While many of these studies use NMDAR antagonists, none of them used KynA. Exercise increases bone density (143) through well-studied mechano-sensing pathways (157), yet the ability of KynA to modulate these benefits remains to be investigated.

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As with the previous vignettes, there are numerous locations where reducing NMDAR activity are congruent with benefits associated with exercise. Antagonizing NMDAR with KynA reduces gut hypermotility and inflammation activation in rat models of colitis (147). Similar results was found using KynA in a dog colon obstruction model (73), though neither study used NMDAR-specific antagonists to confirm NMDA specific targeting. Regardless, exercise is a suggested treatment for irritable bowel syndrome (IBS) (68). NMDARs are involved in hyperalgesia, and inflammation induced pain. KynA reduces the mechanical pain threshold in a rat models of inflammation joint pain (86) and MK-801 improves the pain threshold in a hind-paw inflammation model (27). Exercise potentially decreases pain severity and physical function with minimal adverse effects (77). Ultimately, a link between muscle-derived KynA and modulation of NMDARs in any peripheral tissue will open a large field of study.

G-protein-coupled receptor 35 (GPR35):

GPR35 is a 34 kDa membrane protein discovered in 1997 (96) that is preferentially coupled with $G\alpha_{i/o}$ proteins and at some extent with $G\alpha_{13,}$ $G\alpha_{q/11}$ and, when stimulated, β-arrestin (24, 33, 67, 80). GPR35 is primarily expressed in the gastrointestinal system and in specific immune cells, but also in other cell populations such as small dorsal root ganglia neurons, discrete regions of the central nervous system (CNS), vascular smooth muscle cells, endothelial cells, and adipocytes (21, 25, 33, 85, 114, 153). GPR35 was considered an orphan receptor until KynA was shown to act as an endogenous ligand (153). Since then, other molecules with pharmacological interest were reported to be putative ligands of GPR35 protein, such as NPPB (5-nitro-2-(3-phenylpropylamino)benzoic acid), Zaprinast, lysophosphatidic acid and the widely used drug excipient pamoic acid (140, 141, 159). It is still debated if GPR35 is a receptor for CXCL17.

Activation of GPR35 has been linked to various biological outputs such as modulation of inflammatory pain (21), blood pressure regulation (25, 92), attenuation of immune response (reviewed by: (24)), and, as we mentioned above, adipocyte energy expenditure (3). Associative studies have linked GPR35 activity to certain gastrointestinal disorders, namely inflammatory bowel disease and ulcerative colitis (30, 64), although the exact role of the receptor remains to be elucidated. Olenchock and coworkers (99) have shown that KynA exerts cardioprotective effects in a remote ischemic preconditioning mouse model. Although the mechanism by which KynA protects the heart was not unraveled, we could speculate that GRP35 is a good candidate. GPR35 is expressed in mouse cardiomyocytes and its level is increased upon hypoxia exposure/HIF-1α activation, a typical response of ischemic tissues (117). The ischemic preconditioning model consists in brief periods of sublethal ischemia which confer resistance to a following, more severe, ischemia-reperfusion insult. It is conceivable that during these sublethal events GPR35 expression is increased by the activity of HIF-1α, creating the molecular sensor for the cardioprotective

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activity of KynA. However, the function of GPR35 in cardiac muscle cells is unclear. Adenoviral overexpression of GPR35 causes hypertrophic-like changes in the cell morphology and reduced cell viability and high expression of the receptor is associated with human heart failure (92). On the other side, GPR35 knockout mice have increased systemic blood pressure. Further research needs to be done to clarify basal and stimulated effects of GPR35 in the heart.

A further aspect to consider is the effect of exercise-induced KynA on immune cells. GPR35 expression is high in human neutrophils, T cells, CD14+ monocytes, and dendritic cells and lower in B cells, eosinophils, basophils, and platelets. GPR35 expression was also detected at the protein level in iNKT cells (33). It is tempting to imagine that KynA and GPR35 could be important in muscle to immune cell communication in the context of exercise.

Aryl hydrocarbon receptor (AhR):

AhR is a ligand-activated transcription factor that acts as an integration hub for environmental, dietary and endogenous cues. Therefore, AhR transcriptional activity elicits tissue, context, and ligand-specific responses. AhR belongs to the superfamily of basic helix-loop-helix/PER-ARNT-SIM transcription factors and once activated, binds to xenobiotic-response elements (XRE) present on target genes (54, 74, 94, 119). Among the genes regulated by AhR, the family 1 of Cytochrome P450 genes (e.g. CYP1A1, CYP1A2 and CYP1B1) is often used as markers of AhR activation. AhR controls the expression of genes involved in complex processes such as cell proliferation (pro or anti-, depending on the cell type), inflammation, cell adhesion and cell migration (see (74)). Interestingly, the expression of IL-6 was also seen to be partially regulated by AhR in MCF-7 breast cancer cells (60). However, little is known about AhR activation and its effects in skeletal muscle. High-fat diet-fed AhR-deficient mice have been reported to have a leaner phenotype and better insulin sensitivity compared to wild-type littermate controls. The effect was attributed to increased metabolism in brown adipose tissue and skeletal muscle, but the exact contribution of these tissues in this process is not clear (158).

AhR can be activated by Kyn, KynA, and cinnabarinic acid (CA), although in many cases their endogenous concentrations would be too low (**Table 1**). Of the whole set of Trp metabolites, CA is the one that most resembles the canonical AhR agonist 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD). However, little is known about variation of plasma levels of CA after exercise or in pathological conditions. CA is formed from the dimerization of 3-HAA in oxidative conditions (**Table 1**). Acute bouts of exercise increase ROS production, and in presence of 3-HAA this could lead to CA formation and local AhR activation. On the other hand, activation of the AhR through exercise-induced KynA could be more than a possibility. HepG2, a human hepatic cell line, responded to 100 nM KynA treatment by upregulating the AhR target gene CYP1A1. Yet primary human hepatocytes responded only to 1 to 10 µM KynA (23), posing the question of whether KynA activation of the AhR

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receptor in the liver can occur at physiological levels of KynA, either induced by exercise or a pathological condition. Other studies have linked specific biological activity of AhR activation to KynA, such as TNF-stimulated gene 6 (TSG-6) release by mesenchymal stem cells (50 – 200 μM KynA, (152)) or immune system modulation (reviewed in ref. (154)). Interestingly, exercise is known to ameliorate chronic inflammation (122, 130) but to date we do not know if this effect is at least in part mediated by AhR activation through KynA or CA. Finally, most studies that investigated AhR activation by Trp metabolites used cell-based readouts. Since many cells are able to partially metabolize Trp, it remains to be established which metabolites are actual AhR ligands.

CONCLUSIONS

The catabolism of Trp through the kynurenine pathway generates several metabolites with biological activity. Exercise alters skeletal muscle kynurenine metabolism and can significantly change the levels of some of these metabolites both in the periphery and in the central nervous system. As a consequence, exercise-induced changes in muscle kynurenine metabolism elicit adaptations in several tissues adipose tissue, immune system, and the central nervous system. An added benefit of this metabolic shift in skeletal muscle is improved energy efficiency. It is likely that additional perks could stem from decreased Kyn/3HK and elevated KynA/XA levels (e.g. local antioxidant effects). QA or 3-HAA could have detrimental effects on muscle function, but their intramuscular concentrations following exercise are unknown. Again, due to unchanged KMO and KynU, Kyn metabolism is likely shifted away from 3-HK, 3-HAA, and QA. Therefore, these metabolites were intentionally not discussed here. Of note, it will be important to study the effects of resistance exercise on the kynurenine pathway and how they differ from endurance exercise. In addition to exercise, further studies are needed to explore the roles of KPM under other energetically challenging conditions. This includes both physiological and pathophysiological conditions such as regulation of insulin secretion and fasting responses, or obesity and diabetes, respectively. Understanding how tissues regulate KPM levels, and the biological activities these metabolites can have in a context-specific way, is helping us build a new network of inter-organ communication. As the interest in the peripheral effects of kynurenines grows, it will be exciting to see what other biological roles are revealed for this remarkable collection of Trp metabolites.

List of abbreviations, acronyms, and initialisms:

AA anthranilic acid

AhR aryl hydrocarbon receptor

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AMPA α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid

AMPAR α-amino-3-hydroxy-5-methyl-4-isoxazolepropionic acid receptor

BAIBA β-Aminoisobutyric acid

BBB blood brain barrier

CA cinnabarinic acid

CNS central nervous system

FAD flavin adenine dinucleotide

GPR35 G-protein-coupled receptor 35

HIF-1α hypoxia-inducible factor-1α

IDO indoleamine 2,3-dioxygenase

KATs kynurenine aminotransferases

KMO kynurenine 3-monooxygenase

KPMs kynurenine pathway metabolites

Kyn kynurenine

KynA Kynurenic acid

KynU kynureninase

LATs large neutral amino acid transporters

MAS malate-aspartate shuttle

MDH malate dehydrogenases

NAD^+ nicotinamide adenine dinucleotide

$NADP^+$ nicotinamide adenine dinucleotide phosphate

NMDA *N*-methyl-D-aspartate

NMDAR *N*-methyl-D-aspartate receptor

NPPB 5-nitro-2-(3-phenylpropylamino)benzoic acid

PGCs Peroxisome Proliferator-Activated Receptor γ coactivators

PHDs prolyl-hydroxylases

PicA picolinic acid

PLP pyridoxal-5'-phosphate

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PRC PGC-related coactivator

QA quinolinic acid

ROS reactive oxygen species

TCDD 2,3,7,8-tetrachlorodibenzo-p-dioxin

TDO tryptophan 2,3-dioxygenase

Trp tryptophan

TSG-6 TNF-stimulated gene 6

XA xanthurenic acid

XRE xenobiotic-response elements

3-HAA 3-hydroxyanthranilic acid

3-HAO 3-hydroxyanthranilate 3,4-dioxygenase

3-HK 3-hydroxykynurenine

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Author Contributions

K.M. prepared figures; K.M., M.A., and J.L.R. drafted manuscript; K.M., M.A., and J.L.R. edited and revised manuscript; K.M., M.A., and J.L.R. approved final version of manuscript.

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558 **References**

559 1. **Agudelo LZ**, **Femenía T**, **Orhan F**, **Porsmyr-Palmertz M**, **Goiny M**, **Martinez-Redondo**
560 **V**, **Correia JC**, **Izadi M**, **Bhat M**, **Schuppe-Koistinen I**, **Pettersson AT**, **Ferreira DMS**,
561 **Krook A**, **Barres R**, **Zierath JR**, **Erhardt S**, **Lindskog M**, **Ruas JL**. Skeletal muscle PGC-
562 1α1 modulates kynurenine metabolism and mediates resilience to stress-induced
563 depression. *Cell* 159: 33–45, 2014.

564 2. **Agudelo LZ**, **Ferreira DMS**, **Dadvar S**, **Cervenka I**, **Ketscher L**, **Izadi M**, **Zhengye L**,
565 **Furrer R**, **Handschin C**, **Venckunas T**, **Brazaitis M**, **Kamandulis S**, **Lanner JT**, **Ruas JL**.
566 Skeletal muscle PGC-1α1 reroutes kynurenine metabolism to increase energy
567 efficiency and fatigue-resistance. *Nat Commun* 10: 1–12, 2019.

568 3. **Agudelo LZ**, **Ferreira DMSS**, **Cervenka I**, **Bryzgalova G**, **Dadvar S**, **Jannig PR**,
569 **Pettersson-Klein AT**, **Lakshmikanth T**, **Sustarsic EG**, **Porsmyr-Palmertz M**, **Correia JC**,
570 **Izadi M**, **Martínez-Redondo V**, **Ueland PM**, **Midttun Ø**, **Gerhart-Hines Z**, **Brodin P**,
571 **Pereira T**, **Berggren P-OO**, **Ruas JL**, **Martinez-Redondo V**, **Ueland PM**, **Midttun O**,
572 **Gerhart-Hines Z**, **Brodin P**, **Pereira T**, **Berggren P-OO**, **Ruas JL**. Kynurenic Acid and
573 Gpr35 Regulate Adipose Tissue Energy Homeostasis and Inflammation. *Cell Metab* 27:
574 378-392.e5, 2018.

575 4. **Alberati-Giani D**, **Buchli R**, **Malherbe P**, **Broger C**, **Lang G**, **Kohler C**, **Lahm HW**, **Cesura**
576 **AM**. Isolation and expression of a cDNA clone encoding human kynureninase. *Eur J*
577 *Biochem* 239: 460–468, 1996.

578 5. **Amaral M**, **Levy C**, **Heyes DJ**, **Lafite P**, **Outeiro TF**, **Giorgini F**, **Leys D**, **Scrutton NS**.
579 Structural basis of kynurenine 3-monooxygenase inhibition. *Nature* 496: 382–385,
580 2013.

581 6. **Baek JM**, **Kim JY**, **Yoon KH**, **Oh J**, **Lee MS**. Ebselen is a potential anti-osteoporosis
582 agent by suppressing receptor activator of nuclear factor kappa-B ligand-induced
583 osteoclast differentiation In Vitro and lipopolysaccharide-induced inflammatory bone
584 destruction In Vivo. *Int J Biol Sci* 12: 478–488, 2016.

585 7. **Bal NC**, **Maurya SK**, **Sopariwala DH**, **Sahoo SK**, **Gupta SC**, **Shaikh SA**, **Pant M**,
586 **Rowland LA**, **Goonasekera SA**, **Molkentin JD**, **Periasamy M**. Sarcolipin is a newly
587 identified regulator of muscle-based thermogenesis in mammals. *Nat Med* 18: 1575–
588 1579, 2012.

589 8. **Bender DA**, **Wynick D**. Inhibition of kynureninase (L-kynurenine hydrolase, *EC* 3.7.1.3)

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

by oestrone sulphate: an alternative explanation for abnormal results of tryptophan load tests in women receiving oestrogenic steroids. *Br J Nutr* 45: 269–275, 1981.

9. **Birch PJ**, **Grossman CJ**, **Hayes AG**. Kynurenic acid antagonises responses to NMDA via an action at the strychnine-insensitive glycine receptor. *Eur J Pharmacol* 154: 85–87, 1988.

10. **Bonaldo P**, **Sandri M**. Cellular and molecular mechanisms of muscle atrophy. *DMM Dis. Model. Mech.* 6: 25–39, 2013.

11. **Borgenvik M**, **Nordin M**, **Mattsson CM**, **Enqvist JK**, **Blomstrand E**, **Ekblom B**. Alterations in amino acid concentrations in the plasma and muscle in human subjects during 24 h of simulated adventure racing. *Eur J Appl Physiol* 112: 3679–3688, 2012.

12. **Boström P**, **Wu J**, **Jedrychowski MP**, **Korde A**, **Ye L**, **Lo JC**, **Rasbach K a.**, **Boström EA**, **Choi JH**, **Long JZ**, **Kajimura S**, **Zingaretti MC**, **Vind BF**, **Tu H**, **Cinti S**, **Højlund K**, **Gygi SP**, **Spiegelman BM**. A PGC1-α-dependent myokine that drives brown-fat-like development of white fat and thermogenesis. *Nature* 481: 463–468, 2012.

13. **Bozic M**, **Valdivielso JM**. The potential of targeting NMDA receptors outside the CNS. *Expert Opin Ther Targets* 19: 399–413, 2015.

14. **Bryleva EY**, **Brundin L**. Kynurenine pathway metabolites and suicidality. *Neuropharmacology* 112: 324–330, 2017.

15. **Caspersen C**, **Powell K**, **Christenson G**. Physical Activity, Exercise, and Physical Fitness: Definitions and Distinctions for Health-Related Research. *Public Heal Rep* 100: 126–31, 1985.

16. **Cervenka I**, **Agudelo LZ**, **Ruas JL**. Kynurenines: Tryptophan's metabolites in exercise, inflammation, and mental health. *Science (80-)* 357, 2017.

17. **Chenu C**, **Serre CM**, **Raynal C**, **Burt-Pichat B**, **Delmas PD**. Glutamate receptors are expressed by bone cells and are involved in bone resorption. *Bone* 22: 295–299, 1998.

18. **Chinsomboon J**, **Ruas J**, **Gupta RK**, **Thom R**, **Shoag J**, **Rowe GC**, **Sawada N**, **Raghuram S**, **Arany Z**. The transcriptional coactivator PGC-1α mediates exercise-induced angiogenesis in skeletal muscle. *Proc Natl Acad Sci U S A* 106: 21401–21406, 2009.

19. **Chouchani ET**, **Kazak L**, **Spiegelman BM**. New Advances in Adaptive Thermogenesis: UCP1 and Beyond. *Cell Metab* 29: 27–37, 2019.

20. **Correia JC**, **Ferreira DMS**, **Ruas JL**. Intercellular: Local and systemic actions of skeletal muscle PGC-1s. *Trends Endocrinol. Metab.* 26 Elsevier Inc.: 305–314, 2015.

21. **Cosi C**, **Mannaioni G**, **Cozzi A**, **Carlà V**, **Sili M**, **Cavone L**, **Maratea D**, **Moroni F**. G-protein coupled receptor 35 (GPR35) activation and inflammatory pain: Studies on the antinociceptive effects of kynurenic acid and zaprinast. *Neuropharmacology* 60: 1227–1231, 2011.

22. **De Crescenzo F**, **Lennox A**, **Gibson JC**, **Cordey JH**, **Stockton S**, **Cowen PJ**, **Quested DJ**. Melatonin as a treatment for mood disorders: a systematic review. *Acta Psychiatr Scand* 136: 549–558, 2017.

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

23. **DiNatale BC**, **Murray IA**, **Schroeder JC**, **Flaveny CA**, **Lahoti TS**, **Laurenzana EM**, **Omiecinski CJ**, **Perdew GH**. Kynurenic Acid Is a Potent Endogenous Aryl Hydrocarbon Receptor Ligand that Synergistically Induces Interleukin-6 in the Presence of Inflammatory Signaling. *Toxicol Sci* 115: 89–97, 2010.

24. **Divorty N**, **Mackenzie AE**, **Nicklin SA**, **Milligan G**. G protein-coupled receptor 35: An emerging target in inflammatory and cardiovascular disease. *Front Pharmacol* 6: 1–13, 2015.

25. **Divorty N**, **Milligan G**, **Graham D**, **Nicklin SA**. The Orphan Receptor GPR35 Contributes to Angiotensin II-Induced Hypertension and Cardiac Dysfunction in Mice. *Am J Hypertens* 31: 1049–1058, 2018.

26. **Drummond MJ**, **Fry CS**, **Glynn EL**, **Timmerman KL**, **Dickinson JM**, **Walker DK**, **Gundermann DM**, **Volpi E**, **Rasmussen BB**. Skeletal muscle amino acid transporter expression is increased in young and older adults following resistance exercise. *J Appl Physiol* 111: 135–142, 2011.

27. **Du J**, **Zhou S**, **Coggeshall RE**, **Carlton SM**. N-methyl-D-aspartate-induced excitation and sensitization of normal and inflamed nociceptors. *Neuroscience* 118: 547–562, 2003.

28. **Eastman CL**, **Guilarte TR**. The role of hydrogen peroxide in the in vitro cytotoxicity of 3-hydroxykynurenine. *Neurochem Res* 15: 1101–1107, 1990.

29. **Egan B**, **Zierath JR**. Exercise metabolism and the molecular regulation of skeletal muscle adaptation. *Cell Metab* 17: 162–184, 2013.

30. **Ellinghaus D**, **Folseraas T**, **Holm K**, **Ellinghaus E**, **Melum E**, **Balschun T**, **Laerdahl JK**, **Shiryaev A**, **Gotthardt DN**, **Weismüller TJ**, **Schramm C**, **Wittig M**, **Bergquist A**, **Björnsson E**, **Marschall H-U**, **Vatn M**, **Teufel A**, **Rust C**, **Gieger C**, **Wichmann H-E**, **Runz H**, **Sterneck M**, **Rupp C**, **Braun F**, **Weersma RK**, **Wijmenga C**, **Ponsioen CY**, **Mathew CG**, **Rutgeerts P**, **Vermeire S**, **Schrumpf E**, **Hov JR**, **Manns MP**, **Boberg KM**, **Schreiber S**, **Franke A**, **Karlsen TH**. Genome-wide association analysis in Primary sclerosing cholangitis and ulcerative colitis identifies risk loci at GPR35 and TCF4. *Hepatology* 58: 1074–1083, 2013.

31. **Emet M**, **Ozcan H**, **Ozel L**, **Yayla M**, **Halici Z**, **Hacimuftuoglu A**. A Review of Melatonin, Its Receptors and Drugs. *Eurasian J Med* 48: 135–141, 2016.

32. **Eriksson KF**, **Lindgärde F**. Prevention of Type 2 (non-insulin-dependent) diabetes mellitus by diet and physical exercise The 6-year Malmö feasibility study. *Diabetologia* 34: 891–898, 1991.

33. **Fallarini S**, **Magliulo L**, **Paoletti T**, **de Lalla C**, **Lombardi G**. Expression of functional GPR35 in human iNKT cells. *Biochem Biophys Res Commun* 398: 420–425, 2010.

34. **Fallarino F**, **Grohmann U**, **Vacca C**, **Bianchi R**, **Orabona C**, **Spreca A**, **Fioretti MC**, **Puccetti P**. T cell apoptosis by tryptophan catabolism. *Cell Death Differ* 9: 1069–1077, 2002.

35. **Fazio F**, **Lionetto L**, **Curto M**, **Iacovelli L**, **Cavallari M**, **Zappulla C**, **Ulivieri M**, **Napoletano F**, **Capi M**, **Corigliano V**, **Scaccianoce S**, **Caruso A**, **Miele J**, **De Fusco A**, **Di**

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

Menna L, **Comparelli A**, **De Carolis A**, **Gradini R**, **Nisticò R**, **De Blasi A**, **Girardi P**, **Bruno V**, **Battaglia G**, **Nicoletti F**, **Simmaco M**. Xanthurenic Acid Activates mGlu2/3 metabotropic glutamate receptors and is a potential trait marker for schizophrenia. *Sci Rep* 5, 2015.

36. **Fazio F**, **Lionetto L**, **Curto M**, **Iacovelli L**, **Copeland CS**, **Neale SA**, **Bruno V**, **Battaglia G**, **Salt TE**, **Nicoletti F**. Cinnabarinic acid and xanthurenic acid: Two kynurenine metabolites that interact with metabotropic glutamate receptors. *Neuropharmacology* 112 Elsevier Ltd: 365–372, 2017.

37. **Fazio F**, **Lionetto L**, **Molinaro G**, **Bertrand HO**, **Acher F**, **Ngomba RT**, **Notartomaso S**, **Curini M**, **Rosati O**, **Scarselli P**, **Di Marco R**, **Battaglia G**, **Bruno V**, **Simmaco M**, **Pin JP**, **Nicoletti F**, **Goudet C**. Cinnabarinic acid, an endogenous metabolite of the kynurenine pathway, activates type 4 metabotropic glutamate receptors. *Mol Pharmacol* 81: 643–656, 2012.

38. **Ferreira FS**, **Biasibetti-Brendler H**, **Pierozan P**, **Schmitz F**, **Bertó CG**, **Prezzi CA**, **Manfredini V**, **Wyse ATS**. Kynurenic Acid Restores Nrf2 Levels and Prevents Quinolinic Acid-Induced Toxicity in Rat Striatal Slices. *Mol Neurobiol* 55: 8538–8549, 2018.

39. **Finck BN**, **Kelly DP**. PGC-1 coactivators: Inducible regulators of energy metabolism in health and disease. *J. Clin. Invest.* 116: 615–622, 2006.

40. **Fotiadis D**, **Kanai Y**, **Palacín M**. The SLC3 and SLC7 families of amino acid transporters. *Mol Aspects Med* 34: 139–158, 2013.

41. **Fürst P**. Intracellular muscle free amino acids—their measurement and function. *Proc Nutr Soc* 42: 451–462, 1983.

42. **Gao J**, **Yao L**, **Xia T**, **Liao X**, **Zhu D**, **Xiang Y**. Biochemistry and structural studies of kynurenine 3-monooxygenase reveal allosteric inhibition by Ro 61-8048. *FASEB J* 32: 2036–2045, 2018.

43. **Goldstein LE**, **Leopold MC**, **Huang X**, **Atwood CS**, **Saunders AJ**, **Hartshorn M**, **Lim JT**, **Faget KY**, **Muffat JA**, **Scarpa RC**, **Chylack LT**, **Bowden EF**, **Tanzi RE**, **Bush AI**. 3-hydroxykynurenine and 3-hydroxyanthranilic acid generate hydrogen peroxide and promote α-crystallin cross-linking by metal ion reduction. *Biochemistry* 39: 7266–7275, 2000.

44. **Gould DW**, **Lahart I**, **Carmichael AR**, **Koutedakis Y**, **Metsios GS**. Cancer cachexia prevention via physical exercise: Molecular mechanisms. *J. Cachexia. Sarcopenia Muscle* 4 Wiley Online Library: 111–124, 2013.

45. **Graham TE**, **Sgro V**, **Friars D**, **Gibala MJ**. Glutamate ingestion: The plasma and muscle free amino acid pools of resting humans. *Am J Physiol - Endocrinol Metab* 278: 83–89, 2000.

46. **Hadj-Saad F**, **Lhuissier M**, **Guilland JC**. Effects of acute, submaximal exercise on vitamin B6 metabolism in the rat. *Nutr Res* 15: 1181–1189, 1995.

47. **Han Q**, **Cai T**, **Tagle DA**, **Li J**. Structure, expression, and function of kynurenine aminotransferases in human and rodent brains. *Cell Mol Life Sci* 67: 353–368, 2010.

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

48. **Handschin C**, **Shulman GI**, **Bruce M**, **Handschin C**, **Choi CS**, **Chin S**, **Kim S**, **Kawamori D**, **Kurpad AJ**, **Neubauer N**, **Hu J**, **Mootha VK**. Abnormal glucose homeostasis in skeletal muscle-specific PGC-1alpha knockout mice reveals skeletal muscle-pancreatic beta cell crosstalk. *J Clin Invest* 117: 3463–3474, 2007.

49. **Hansen KB**, **Yi F**, **Perszyk RE**, **Furukawa H**, **Wollmuth LP**, **Gibb AJ**, **Traynelis SF**. Structure, function, and allosteric modulation of NMDA receptors. *J Gen Physiol* 150: 1081–1105, 2018.

50. **Hatazawa Y**, **Qian K**, **Gong DW**, **Kamei Y**. PGC-1α regulates alanine metabolism in muscle cells. *PLoS One* 13: 1–13, 2018.

51. **Hatazawa Y**, **Senoo N**, **Tadaishi M**, **Ogawa Y**, **Ezaki O**, **Kamei Y**, **Miura S**. Metabolomic analysis of the skeletal muscle mice overexpressing PGC-1α. *PLoS One* 10: 1–17, 2015.

52. **Hatazawa Y**, **Tadaishi M**, **Nagaike Y**, **Morita A**, **Ogawa Y**, **Ezaki O**, **Takai-Igarashi T**, **Kitaura Y**, **Shimomura Y**, **Kamei Y**, **Miura S**. PGC-1α-mediated branched-chain amino acid metabolism in the skeletal muscle. *PLoS One* 9: 1–10, 2014.

53. **Hawley JA**, **Hargreaves M**, **Joyner MJ**, **Zierath JR**. Integrative biology of exercise. *Cell* 159: 738–749, 2014.

54. **Hidaka T**, **Fujimura T**, **Aiba S**. Aryl Hydrocarbon Receptor Modulates Carcinogenesis and Maintenance of Skin Cancers. *Front. Med.* 6: 194, 2019.

55. **Hinoi E**, **Fujimori S**, **Yoneda Y**. Modulation of cellular differentiation by N-methyl-D-aspartate receptors in osteoblasts. *FASEB J* 17: 1532–1534, 2003.

56. **Hiramatsu R**, **Hara T**, **Akimoto H**, **Takikawa O**, **Kawabe T**, **Isobe K**, **Nagase F**. Cinnabarinic acid generated from 3-hydroxyanthranilic acid strongly induces apoptosis in thymocytes through the generation of reactive oxygen species and the induction of caspase. *J Cell Biochem* 103: 42–53, 2008.

57. **Ho ML**, **Tsai TN**, **Chang JK**, **Shao TS**, **Jeng YR**, **Hsu C**. Down-regulation of N-methyl D-aspartate receptor in rat-modeled disuse osteopenia. *Osteoporos Int* 16: 1780–1788, 2005.

58. **Hodson N**, **Brown T**, **Joanisse S**, **Aguirre N**, **West DWD**, **Moore DR**, **Baar K**, **Breen L**, **Philp A**. Characterisation of L-type amino acid transporter 1 (LAT1) expression in human skeletal muscle by immunofluorescent microscopy. *Nutrients* 10, 2018.

59. **Hogan-Cann AD**, **Anderson CM**. Physiological Roles of Non-Neuronal NMDA Receptors. *Trends Pharmacol Sci* 37: 750–767, 2016.

60. **Hollingshead BD**, **Beischlag T V**, **Dinatale BC**, **Ramadoss P**, **Perdew GH**. Inflammatory signaling and aryl hydrocarbon receptor mediate synergistic induction of interleukin 6 in MCF-7 cells. *Cancer Res* 68: 3609–3617, 2008.

61. **Hötting K**, **Röder B**. Beneficial effects of physical exercise on neuroplasticity and cognition. *Neurosci. Biobehav. Rev.* 37: 2243–2257, 2013.

62. **Huang XT**, **Li C**, **Peng XP**, **Guo J**, **Yue SJ**, **Liu W**, **Zhao FY**, **Han JZ**, **Huang YH**, **Yang-Li**, **Cheng QM**, **Zhou ZG**, **Chen C**, **Feng DD**, **Luo ZQ**. An excessive increase in glutamate contributes to glucose-toxicity in β-cells via activation of pancreatic NMDA receptors

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

in rodent diabetes. *Sci Rep* 7: 1–14, 2017.

63. **Hutchinson JP**, **Rowland P**, **Taylor MRD**, **Christodoulou EM**, **Haslam C**, **Hobbs CI**, **Holmes DS**, **Homes P**, **Liddle J**, **Mole DJ**, **Uings I**, **Walker AL**, **Webster SP**, **Mowat CG**, **Chung C-W**. Structural and mechanistic basis of differentiated inhibitors of the acute pancreatitis target kynurenine-3-monooxygenase. *Nat Commun* 8: 15827, 2017.

64. **Imielinski M**, **Baldassano RN**, **Griffiths A**, **Russell RK**, **Annese V**, **Dubinsky M**, **Kugathasan S**, **Bradfield JP**, **Walters TD**, **Sleiman P**, **Kim CE**, **Muise A**, **Wang K**, **Glessner JT**, **Saeed S**, **Zhang H**, **Frackelton EC**, **Hou C**, **Flory JH**, **Otieno G**, **Chiavacci RM**, **Grundmeier R**, **Castro M**, **Latiano A**, **Dallapiccola B**, **Stempak J**, **Abrams DJ**, **Taylor K**, **McGovern D**, **Heyman MB**, **Ferry GD**, **Kirschner B**, **Lee J**, **Essers J**, **Grand R**, **Stephens M**, **Levine A**, **Piccoli D**, **Van Limbergen J**, **Cucchiara S**, **Monos DS**, **Guthery SL**, **Denson L**, **Wilson DC**, **Grant SFA**, **Daly M**, **Silverberg MS**, **Satsangi J**, **Hakonarson H**. Common variants at five new loci associated with early-onset inflammatory bowel disease. *Nat Genet* 41: 1335–1340, 2009.

65. **Itzstein C**, **Espinosa L**, **Delmas PD**, **Chenu C**. Specific antagonists of NMDA receptors prevent osteoclast sealing zone formation required for bone resorption. *Biochem Biophys Res Commun* 268: 201–209, 2000.

66. **Janssen I**, **Heymsfield SB**, **Wang Z**, **Ross R**. Skeletal muscle mass and distribution in 468 men and women aged 18-88 yr. *J Appl Physiol* 89: 81–88, 2000.

67. **Jenkins L**, **Brea J**, **Smith NJ**, **Hudson BD**, **Reilly G**, **Bryant NJ**, **Castro M**, **Loza M-I**, **Milligan G**. Identification of novel species-selective agonists of the G-protein-coupled receptor GPR35 that promote recruitment of β-arrestin-2 and activate Gα13. *Biochem J* 432: 451–459, 2010.

68. **Johannesson E**, **Simrén M**, **Strid H**, **Bajor A**, **Sadik R**. Physical activity improves symptoms in irritable bowel syndrome: A randomized controlled trial. *Am J Gastroenterol* 106: 915–922, 2011.

69. **Joshi AD**, **Carter DE**, **Harper TA**, **Elferink CJ**. Aryl hydrocarbon receptor-dependent stanniocalcin 2 induction by cinnabarinic acid provides cytoprotection against endoplasmic reticulum and oxidative stress. *J Pharmacol Exp Ther* 353: 201–212, 2015.

70. **Kaiser H**, **Yu K**, **Pandya C**, **Mendhe B**, **Isales CM**, **McGee-Lawrence ME**, **Johnson M**, **Fulzele S**, **Hamrick MW**. Kynurenine, a Tryptophan Metabolite That Increases with Age, Induces Muscle Atrophy and Lipid Peroxidation. *Oxid Med Cell Longev* 2019, 2019.

71. **Kane DA**. Lactate oxidation at the mitochondria: A lactate-malate-aspartate shuttle at work. *Front Neurosci* 8: 1–6, 2014.

72. **Karstoft K**, **Pedersen BK**. Skeletal muscle as a gene regulatory endocrine organ. *Curr Opin Clin Nutr Metab Care* 19: 270–275, 2016.

73. **Kaszaki J**, **Palásthy Z**, **Érczes D**, **Rácz A**, **Torday C**, **Varga G**, **Vécsei L**, **Boros M**. Kynurenic acid inhibits intestinal hypermotility and xanthine oxidase activity during experimental colon obstruction in dogs. *Neurogastroenterol Motil* 20: 53–62, 2008.

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

74. **Larigot L**, **Juricek L**, **Dairou J**, **Coumoul X**. AhR signaling pathways and regulatory functions. *Biochim Open* 7: 1–9, 2018.

75. **Lin J**, **Wu H**, **Tarr PT**, **Zhang CY**, **Wu Z**, **Boss O**, **Michael LF**, **Puigserver P**, **Isotani E**, **Olson EN**, **Lowell BB**, **Bassel-Duby R**, **Spiegelman BM**. Transcriptional co-activator PGC-1α drives the formation of slow-twitch muscle fibres. *Nature* 418: 797–801, 2002.

76. **Lin TH**, **Yang R Sen**, **Tang CH**, **Wu MY**, **Fu WM**. Regulation of the maturation of osteoblasts and osteoclastogenesis by glutamate. *Eur J Pharmacol* 589: 37–44, 2008.

77. **Lj G**, **Ra M**, **Clarke C**, **Martin D**, **La C**, **Bh S**, **Geneen LJ**, **Moore RA**, **Clarke C**, **Martin D**, **Colvin LA**, **Smith BH**. Physical activity and exercise for chronic pain in adults. *Cochrane database Syst Rev* 1: CD011279, 2017.

78. **Lowe MM**, **Mold JE**, **Kanwar B**, **Huang Y**, **Louie A**, **Pollastri MP**, **Wang C**, **Patel G**, **Franks DG**, **Schlezinger J**, **Sherr DH**, **Silverstone AE**, **Hahn ME**, **McCune JM**. Identification of Cinnabarinic Acid as a Novel Endogenous Aryl Hydrocarbon Receptor Ligand That Drives IL-22 Production. *PLoS One* 9: e87877, 2014.

79. **Lugo-Huitrón R**, **Blanco-Ayala T**, **Ugalde-Muñiz P**, **Carrillo-Mora P**, **Pedraza-Chaverrí J**, **Silva-Adaya D**, **Maldonado PD**, **Torres I**, **Pinzón E**, **Ortiz-Islas E**, **López T**, **García E**, **Pineda B**, **Torres-Ramos M**, **Santamaría A**, **La Cruz VP De**. On the antioxidant properties of kynurenic acid: Free radical scavenging activity and inhibition of oxidative stress. *Neurotoxicol Teratol* 33: 538–547, 2011.

80. **MacKenzie AE**, **Lappin JE**, **Taylor DL**, **Nicklin SA**, **Milligan G**. GPR35 as a novel therapeutic target. *Front Endocrinol (Lausanne)* 2: 1–10, 2011.

81. **Marquard J**, **Otter S**, **Welters A**, **Stirban A**, **Fischer A**, **Eglinger J**, **Herebian D**, **Kletke O**, **Klemen MS**, **Stožer A**, **Wnendt S**, **Piemonti L**, **Köhler M**, **Ferrer J**, **Thorens B**, **Schliess F**, **Rupnik MS**, **Heise T**, **Berggren PO**, **Klöcker N**, **Meissner T**, **Mayatepek E**, **Eberhard D**, **Kragl M**, **Lammert E**. Characterization of pancreatic NMDA receptors as possible drug targets for diabetes treatment. *Nat Med* 21: 363–376, 2015.

82. **Martínez-Redondo V**, **Jannig PR**, **Correia JC**, **Ferreira DMS**, **Cervenka I**, **Lindvall JM**, **Sinha I**, **Izadi M**, **Pettersson-Klein AT**, **Agudelo LZ**, **Gimenez-Cassina A**, **Brum PC**, **Dahlman-Wright K**, **Ruas JL**. Peroxisome proliferator-activated receptor γ coactivator-1 α isoforms selectively regulate multiple splicing events on target genes. *J Biol Chem* 291: 15169–15184, 2016.

83. **Martínez-Redondo V**, **Pettersson AT**, **Ruas JL**. The hitchhiker's guide to PGC-1α isoform structure and biological functions. *Diabetologia* 58: 1969–77, 2015.

84. **Maughan RJ**. Nutrition in Sport. VII. John Wiley & Sons, Ltd, 2000.

85. **McCallum JE**, **Mackenzie AE**, **Divorty N**, **Clarke C**, **Delles C**, **Milligan G**, **Nicklin SA**. G-Protein-Coupled Receptor 35 Mediates Human Saphenous Vein Vascular Smooth Muscle Cell Migration and Endothelial Cell Proliferation. *J Vasc Res* 52: 383–395, 2015.

86. **Mecs L**, **Tuboly G**, **Nagy E**, **Benedek G**, **Horvath G**. The peripheral antinociceptive effects of endomorphin-1 and kynurenic acid in the rat inflamed joint model. *Anesth*

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

Analg 109: 1297–1304, 2009.

87. **Mehta A**, **Prabhakar M**, **Kumar P**, **Deshmukh R**, **Sharma PL**. Excitotoxicity: Bridge to various triggers in neurodegenerative disorders. *Eur J Pharmacol* 698: 6–18, 2013.

88. **Mentaverri R**, **Kamel S**, **Wattel A**, **Prouillet C**, **Sevenet N**, **Petit JP**, **Tordjmann T**, **Brazier M**. Regulation of bone resorption and osteoclast survival by nitric oxide: Possible involvement of NMDA-receptor. *J Cell Biochem* 88: 1145–1156, 2003.

89. **Merk D**, **Gabler M**, **Gomez RC**, **Flesch D**, **Hanke T**, **Kaiser A**, **Lamers C**, **Werz O**, **Schneider G**, **Schubert-Zsilavecz M**. Anthranilic acid derivatives as novel ligands for farnesoid X receptor (FXR). *Bioorganic Med Chem* 22: 2447–2460, 2014.

90. **Mezrich JD**, **Fechner JH**, **Zhang X**, **Johnson BP**, **Burlingham WJ**, **Bradfield CA**. An Interaction between Kynurenine and the Aryl Hydrocarbon Receptor Can Generate Regulatory T Cells. *J Immunol* 185: 3190–3198, 2010.

91. **Mills R**, **Taylor-weiner H**, **Correia JC**, **Agudelo LZ**, **Allodi I**, **Kolonelou C**, **Martinez-redondo V**, **Ferreira DMS**, **Nichterwitz S**, **Comley LH**, **Lundin V**, **Hedlund E**, **Ruas JL**, **Teixeira AI**. Neurturin is a PGC-1 a 1-controlled myokine that promotes motor neuron recruitment and neuromuscular junction formation. *Mol Metab* 7: 12–22, 2018.

92. **Min K-D**, **Asakura M**, **Liao Y**, **Nakamaru K**, **Okazaki H**, **Takahashi T**, **Fujimoto K**, **Ito S**, **Takahashi A**, **Asanuma H**, **Yamazaki S**, **Minamino T**, **Sanada S**, **Seguchi O**, **Nakano A**, **Ando Y**, **Otsuka T**, **Furukawa H**, **Isomura T**, **Takashima S**, **Mochizuki N**, **Kitakaze M**. Identification of genes related to heart failure using global gene expression profiling of human failing myocardium. *Biochem Biophys Res Commun* 393: 55–60, 2010.

93. **Mowat CG**. Role of Kynurenine Pathway in Cancer Biology. In: *Targeting the Broadly Pathogenic Kynurenine Pathway*, edited by Mittal S. Springer International Publishing, p. 273–286.

94. **Mulero-Navarro S**, **Fernandez-Salguero PM**. New trends in Aryl hydrocarbon receptor biology. *Front Cell Dev Biol* 4: 1–14, 2016.

95. **Neale SA**, **Copeland CS**, **Uebele VN**, **Thomson FJ**, **Salt TE**. Modulation of hippocampal synaptic transmission by the kynurenine pathway member xanthurenic acid and other vglut inhibitors. *Neuropsychopharmacology* 38: 1060–1067, 2013.

96. **O'Dowd BF**, **Nguyen T**, **Marchese A**, **Cheng R**, **Lynch KR**, **Heng HHQ**, **Kolakowski LF**, **George SR**. Discovery of three novel G-protein-coupled receptor genes. *Genomics* (1998). doi: 10.1006/geno.1998.5095.

97. **Okamoto H**. Kynurenine 3-Monooxygenase (Hydroxylase) (Rat Liver). *Methods Enzymol* 17: 460–463, 1970.

98. **Okuda S**, **Nishiyama N**, **Saito H**, **Katsuki H**. 3-Hydroxykynurenine, an Endogenous Oxidative Stress Generator, Causes Neuronal Cell Death with Apoptotic Features and Region Selectivity. *J Neurochem* 70: 299–307, 1998.

99. **Olenchock BA**, **Moslehi J**, **Baik AH**, **Davidson SM**, **Williams J**, **Gibson WJ**, **Pierce KA**, **Miller CM**, **Hanse EA**, **Kelekar A**, **Sullivan LB**, **Wagers AJ**, **Clish CB**, **Vander Heiden MG**, **Kaelin WG**. EGLN1 Inhibition and Rerouting of α-Ketoglutarate Suffice for

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

873 Remote Ischemic Protection. *Cell* 164: 884–895, 2016.

874 100. **Opitz CA**, **Litzenburger UM**, **Sahm F**, **Ott M**, **Tritschler I**, **Trump S**, **Schumacher T**,
875 **Jestaedt L**, **Schrenk D**, **Weller M**, **Jugold M**, **Guillemin GJ**, **Miller CL**, **Lutz C**,
876 **Radlwimmer B**, **Lehmann I**, **Von Deimling A**, **Wick W**, **Platten M**. An endogenous
877 tumour-promoting ligand of the human aryl hydrocarbon receptor. *Nature* 478: 197–
878 203, 2011.

879 101. **Ostrowski K**, **Rohde T**, **Zacho M**, **Asp S**, **Pedersen BK**. Evidence that interleukin-6 is
880 produced in human skeletal muscle during prolonged running. *J Physiol* 508: 949–953,
881 1998.

882 102. **Overmyer KA**, **Evans CR**, **Qi NR**, **Minogue CE**, **Carson JJ**, **Chermside-Scabbo CJ**, **Koch**
883 **LG**, **Britton SL**, **Pagliarini DJ**, **Coon JJ**, **Burant CF**. Maximal oxidative capacity during
884 exercise is associated with skeletal muscle fuel selection and dynamic changes in
885 mitochondrial protein acetylation. *Cell Metab* 21: 468–478, 2015.

886 103. **Pawlak D**, **Tankiewicz A**, **Matys T**, **Buczko W**. Peripheral distribution of kynurenine
887 metabolites and activity of kynurenine pathway enzymes in renal failure. *J Physiol*
888 *Pharmacol* 54: 175–189, 2003.

889 104. **Peet NM**, **Grabowski PS**, **Laketić-Ljubojević I**, **Skerry TM**. The glutamate receptor
890 antagonist MK801 modulates bone resorption in vitro by a mechanism predominantly
891 involving osteoclast differentiation. *FASEB J* 13: 2179–2185, 1999.

892 105. **Pérez-González A**, **Alvarez-Idaboy JR**, **Galano A**. Free-radical scavenging by
893 tryptophan and its metabolites through electron transfer based processes. *J Mol*
894 *Model* 21, 2015.

895 106. **Pettersson-Klein AT**, **Izadi M**, **Ferreira DMS**, **Cervenka I**, **Correia JC**, **Martinez-**
896 **Redondo V**, **Southern M**, **Cameron M**, **Kamenecka T**, **Agudelo LZ**, **Porsmyr-Palmertz**
897 **M**, **Martens U**, **Lundgren B**, **Otrocka M**, **Jenmalm-Jensen A**, **Griffin PR**, **Ruas JL**. Small
898 molecule PGC-1α1 protein stabilizers induce adipocyte Ucp1 expression and
899 uncoupled mitochondrial respiration. *Mol Metab* 9: 28–42, 2018.

900 107. **Phu S**, **Boersma D**, **Duque G**. Exercise and Sarcopenia. *J Clin Densitom* 18: 488–492,
901 2015.

902 108. **Pillon NJ**, **Gabriel BM**, **Dollet L**, **Smith JAB**, **Sardón Puig L**, **Botella J**, **Bishop DJ**, **Krook**
903 **A**, **Zierath JR**. Transcriptomic profiling of skeletal muscle adaptations to exercise and
904 inactivity. *Nat Commun* 11, 2020.

905 109. **Prescott C**, **Weeks AM**, **Staley KJ**, **Partin KM**. Kynurenic acid has a dual action on
906 AMPA receptor responses. *Neurosci Lett* 402: 108–112, 2006.

907 110. **Puigserver P**, **Wu Z**, **Park CW**, **Graves R**, **Wright M**, **Spiegelman BM**. A Cold-Inducible
908 Coactivator of Nuclear Receptors Linked to Adaptive Thermogenesis. *Cell* 92: 829–
909 839, 1998.

910 111. **Quagliariello E**, **Papa S**, **Saccone C**, **Alifano A**. Effect of 3-hydroxyanthranilic acid on
911 the mitochondrial respiratory system. *Biochem J* 91: 137–146, 1964.

912 112. **Raghavan S**, **Tria GS**, **Shen HC**, **Ding FX**, **Taggart AK**, **Ren N**, **Wilsie LC**, **Krsmanovic ML**,

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

Holt TG, **Wolff MS**, **Waters MG**, **Hammond ML**, **Tata JR**, **Colletti SL**. Tetrahydro anthranilic acid as a surrogate for anthranilic acid: Application to the discovery of potent niacin receptor agonists. *Bioorganic Med Chem Lett* 18: 3163–3167, 2008.

113. **Rao RR**, **Long JZ**, **White JP**, **Svensson KJ**, **Lou J**, **Lokurkar I**, **Jedrychowski MP**, **Ruas JL**, **Wrann CD**, **Lo JC**, **Camera DM**, **Lachey J**, **Gygi S**, **Seehra J**, **Hawley JA**, **Spiegelman BM**. Meteorin-like is a hormone that regulates immune-adipose interactions to increase beige fat thermogenesis. *Cell* 157: 1279–1291, 2014.

114. **Resta F**, **Masi A**, **Sili M**, **Laurino A**, **Moroni F**, **Mannaioni G**. Kynurenic acid and zaprinast induce analgesia by modulating HCN channels through GPR35 activation. *Neuropharmacology* 108: 136–143, 2016.

115. **Richter EA**, **Hargreaves M**. Exercise, GLUT4, and skeletal muscle glucose uptake. *Physiol Rev* 93: 993–1017, 2013.

116. **Roberts LD**, **Boström P**, **O'Sullivan JF**, **Schinzel RT**, **Lewis GD**, **Dejam A**, **Lee Y-K**, **Palma MJ**, **Calhoun S**, **Georgiadi A**, **Chen M-H**, **Ramachandran VS**, **Larson MG**, **Bouchard C**, **Rankinen T**, **Souza AL**, **Clish CB**, **Wang TJ**, **Estall JL**, **Soukas AA**, **Cowan CA**, **Spiegelman BM**, **Gerszten RE**. β-Aminoisobutyric Acid Induces Browning of White Fat and Hepatic β-Oxidation and Is Inversely Correlated with Cardiometabolic Risk Factors. *Cell Metab* 19: 96–108, 2014.

117. **Ronkainen VP**, **Tuomainen T**, **Huusko J**, **Laidinen S**, **Malinen M**, **Palvimo JJ**, **Ylä-Herttuala S**, **Vuolteenaho O**, **Tavi P**. Hypoxia-inducible factor 1-induced G protein-coupled receptor 35 expression is an early marker of progressive cardiac remodelling. *Cardiovasc Res* 101: 69–77, 2014.

118. **Rossi F**, **Miggiano R**, **Ferraris DM**, **Rizzi M**. The synthesis of kynurenic acid in mammals: An updated kynurenine aminotransferase structural KATalogue. *Front Mol Biosci* 6: 1–9, 2019.

119. **Rothhammer V**, **Quintana FJ**. The aryl hydrocarbon receptor: an environmental sensor integrating immune responses in health and disease. *Nat Rev Immunol* 19: 184–197, 2019.

120. **Rowlands DS**, **Nelson AR**, **Raymond F**, **Metairon S**, **Mansourian R**, **Clarke J**, **Stellingwerff T**, **Phillips SM**. Protein-leucine ingestion activates a regenerative inflammo-myogenic transcriptome in skeletal muscle following intense endurance exercise. *Physiol Genomics* 48: 21–32, 2016.

121. **Ruas JLL**, **White JPP**, **Rao RRR**, **Kleiner S**, **Brannan KTT**, **Harrison BCC**, **Greene NPP**, **Wu J**, **Estall JLL**, **Irving BAA**, **Lanza IRR**, **Rasbach KAA**, **Okutsu M**, **Nair KSS**, **Yan Z**, **Leinwand LAA**, **Spiegelman BMM**. A PGC-1a isoform induced by resistance training regulates skeletal muscle hypertrophy. *Cell* 151: 1319–1331, 2012.

122. **Sallam N**, **Laher I**. Exercise modulates oxidative stress and inflammation in aging and cardiovascular diseases. *Oxid Med Cell Longev* 2016: 46–54, 2016.

123. **Sathyasaikumar K V.**, **Tararina M**, **Wu HQ**, **Neale SA**, **Weisz F**, **Salt TE**, **Schwarcz R**. Xanthurenic Acid Formation from 3-Hydroxykynurenine in the Mammalian Brain: Neurochemical Characterization and Physiological Effects. *Neuroscience* 367: 85–97,

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

954 2017.

955 124. **Schantz PG**, **Henriksson J**. Enzyme levels of the NADH shuttle systems:
956 Measurements in isolated muscle fibres from humans of differing physical activity.
957 *Acta Physiol Scand* 129: 505–515, 1987.

958 125. **Schantz PG**, **Sjoberg B**, **Svedenhag J**. Malate-aspartate and alpha-glycerophosphate
959 shuttle enzyme levels in human skeletal muscle: Methodological considerations and
960 effect of endurance training. *Acta Physiol Scand* 128: 397–407, 1986.

961 126. **Schiaffino S**, **Reggiani C**. Fiber Types in Mammalian Skeletal Muscles. *Physiol Rev* 91:
962 1447–1531, 2011.

963 127. **Schlittler M**, **Goiny M**, **Agudelo LZ**, **Venckunas T**, **Brazaitis M**, **Skurvydas A**,
964 **Kamandulis S**, **Ruas JL**, **Erhardt S**, **Westerblad H**, **Andersson DC**. Endurance exercise
965 increases skeletal muscle kynurenine aminotransferases and plasma kynurenic acid in
966 humans. *Am J Physiol - Cell Physiol* 310: C836–C840, 2016.

967 128. **Sekine A**, **Kuroki Y**, **Urata T**, **Mori N**, **Fukuwatari T**. Inhibition of Large Neutral Amino
968 Acid Transporters Suppresses Kynurenic Acid Production Via Inhibition of Kynurenine
969 Uptake in Rodent Brain. *Neurochem Res* 41: 2256–2266, 2016.

970 129. **Sekine A**, **Okamoto M**, **Kanatani Y**, **Sano M**, **Shibata K**, **Fukuwatari T**. Amino acids
971 inhibit kynurenic acid formation via suppression of kynurenine uptake or kynurenic
972 acid synthesis in rat brain in vitro. *Springerplus* 4, 2015.

973 130. **Shaw DM**, **Merien F**, **Braakhuis A**, **Dulson D**. T-cells and their cytokine production:
974 The anti-inflammatory and immunosuppressive effects of strenuous exercise.
975 *Cytokine* 104 Academic Press: 136–142, 2018.

976 131. **Shi H**, **Enriquez A**, **Rapadas M**, **Martin EMMA**, **Wang R**, **Moreau J**, **Lim CK**, **Szot JO**, **Ip**
977 **E**, **Hughes JN**, **Sugimoto K**, **Humphreys DT**, **McInerney-Leo AM**, **Leo PJ**, **Maghzal GJ**,
978 **Halliday J**, **Smith J**, **Colley A**, **Mark PR**, **Collins F**, **Sillence DO**, **Winlaw DS**, **Ho JWK**,
979 **Guillemin GJ**, **Brown MA**, **Kikuchi K**, **Thomas PQ**, **Stocker R**, **Giannoulatou E**,
980 **Chapman G**, **Duncan EL**, **Sparrow DB**, **Dunwoodie SL**. NAD Deficiency, Congenital
981 Malformations, and Niacin Supplementation. *N Engl J Med* 377: 544–552, 2017.

982 132. **Sigal RJ**, **Kenny GP**, **Wasserman DH**, **Castaneda-Sceppa C**, **White RD**. Physical
983 activity/exercise and type 2 diabetes: A consensus statement from the American
984 Diabetes Association. *Diabetes Care* 29: 1433–1438, 2006.

985 133. **Silveira H**, **Moraes H**, **Oliveira N**, **Coutinho ESF**, **Laks J**, **Deslandes A**. Physical exercise
986 and clinically depressed patients: A systematic review and meta-analysis.
987 *Neuropsychobiology* 67: 61–68, 2013.

988 134. **Steensberg A**, **van Hall G**, **Osada T**, **Sacchetti M**, **Saltin B**, **Pedersen BK**. Production of
989 interleukin-6 in contracting human skeletal muscles can account for the exercise-
990 induced increase in plasma interleukin-6. *J Physiol* 529: 237–242, 2000.

991 135. **Stone TW**. Neuropharmacology of quinolinic and kynurenic acids. *Pharmacol Rev* 45:
992 309 LP – 379, 1993.

993 136. **Stone TW**. Kynurenines in the CNS: From endogenous obscurity to therapeutic

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

importance. *Prog. Neurobiol.* 64: 185–218, 2001.

137. **Stone TW**, **Addae JI**. The pharmacological manipulation of glutamate receptors and neuroprotection. *Eur J Pharmacol* 447: 285–296, 2002.

138. **Stone TW**, **Forrest CM**, **Darlington LG**. Kynurenine pathway inhibition as a therapeutic strategy for neuroprotection. *FEBS J* 279: 1386–1397, 2012.

139. **Takeda M**, **Noshiro R**, **Onozato ML**, **Tojo A**, **Hasannejad H**, **Huang XL**, **Narikawa S**, **Endou H**. Evidence for a role of human organic anion transporters in the muscular side effects of HMG-CoA reductase inhibitors. *Eur J Pharmacol* 483: 133–138, 2004.

140. **Taniguchi Y**, **Tonai-Kachi H**, **Shinjo K**. Zaprinast, a well-known cyclic guanosine monophosphate-specific phosphodiesterase inhibitor, is an agonist for GPR35. *FEBS Lett* 580: 5003–5008, 2006.

141. **Taniguchi Y**, **Tonai-Kachi H**, **Shinjo K**. 5-Nitro-2-(3-phenylpropylamino)benzoic acid is a GPR35 agonist. *Pharmacology* 82: 245–249, 2008.

142. **Theofylaktopoulou D**, **Midttun O**, **Ulvik A**, **Ueland PM**, **Tell GS**, **Vollset SE**, **Nygard O**, **Eussen SJPM**. A community-based study on determinants of circulating markers of cellular immune activation and kynurenines: the Hordaland Health Study. *Clin Exp Immunol* 173: 121–130, 2013.

143. **Turner CH**, **Robling AG**. Mechanisms by which exercise improves bone strength. *J Bone Miner Metab* 23: 16–22, 2005.

144. **Uwai Y**, **Honjo E**. Transport of xanthurenic acid by rat/human organic anion transporters OAT1 and OAT3. *Biosci Biotechnol Biochem* 77: 1517–1521, 2013.

145. **Uwai Y**, **Honjo H**, **Iwamoto K**. Interaction and transport of kynurenic acid via human organic anion transporters hOAT1 and hOAT3. *Pharmacol Res* 65: 254–260, 2012.

146. **Vamos E**, **Pardutz A**, **Klivenyi P**, **Toldi J**, **Vecsei L**. The role of kynurenines in disorders of the central nervous system: Possibilities for neuroprotection. *J Neurol Sci* 283: 21–27, 2009.

147. **Varga G**, **Érces D**, **Fazekas B**, **Fülöp M**, **Kovács T**, **Kaszaki J**, **Fülöp F**, **Vécsei L**, **Boros M**. N-Methyl-d-aspartate receptor antagonism decreases motility and inflammatory activation in the early phase of acute experimental colitis in the rat. *Neurogastroenterol Motil* 22: 7–9, 2010.

148. **Varnavas A**, **Lassiani L**, **Valenta V**, **Berti F**, **Mennuni L**, **Makovec F**. Anthranilic acid derivatives: a new class of non-peptide CCK1 receptor antagonists. *Bioorg Med Chem* 11: 741–51, 2003.

149. **Vécsei L**, **Szalárdy L**, **Fülöp F**, **Toldi J**. Kynurenines in the CNS: Recent advances and new questions. *Nat. Rev. Drug Discov.* 12: 64–82, 2013.

150. **Wallberg-Henriksson H**, **Zierath JR**. Metabolism: Exercise remodels subcutaneous fat tissue and improves metabolism. *Nat. Rev. Endocrinol.* 11 Nature Publishing Group: 198–200, 2015.

151. **Walsh HA**, **Botting NP**. Purification and biochemical characterization of some of the

Downloaded from journals.physiology.org/journal/ajpcell at Dalhousie Univ DAL 11762 (129.173.072.087) on March 26, 2020.

properties of recombinant human kynureninase. *Eur J Biochem* 269: 2069–2074, 2002.

152. **Wang G**, **Cao K**, **Liu K**, **Xue Y**, **Roberts AI**, **Li F**, **Han Y**, **Rabson AB**, **Wang Y**, **Shi Y**. Kynurenic acid, an IDO metabolite, controls TSG-6-mediated immunosuppression of human mesenchymal stem cells. *Cell Death Differ* 25: 1209–1223, 2018.

153. **Wang J**, **Simonavicius N**, **Wu X**, **Swaminath G**, **Reagan J**, **Tian H**, **Ling L**. Kynurenic acid as a ligand for orphan G protein-coupled receptor GPR35. *J Biol Chem* 281: 22021–22028, 2006.

154. **Wirthgen E**, **Hoeflich A**, **Rebl A**, **Günther J**. Kynurenic Acid: The Janus-faced role of an immunomodulatory tryptophan metabolite and its link to pathological conditions. *Front Immunol* 8, 2018.

155. **Wolin KY**, **Yan Y**, **Colditz GA**, **Lee IM**. Physical activity and colon cancer prevention: A meta-analysis. *Br J Cancer* 100: 611–616, 2009.

156. **Wrann CD**, **White JP**, **Salogiannnis J**, **Laznik-Bogoslavski D**, **Wu J**, **Ma D**, **Lin JD**, **Greenberg ME**, **Spiegelman BM**. Exercise Induces Hippocampal BDNF through a PGC-1α/FNDC5 Pathway. *Cell Metab* 18: 649–659, 2013.

157. **Xiao Z**, **Quarles LD**. Physiological mechanisms and therapeutic potential of bone mechanosensing. *Rev Endocr Metab Disord* 16: 115–129, 2015.

158. **Xu C-X**, **Wang C**, **Zhang Z-M**, **Jaeger CD**, **Krager SL**, **Bottum KM**, **Liu J**, **Liao D-F**, **Tischkau SA**. Aryl hydrocarbon receptor deficiency protects mice from diet-induced adiposity and metabolic disorders through increased energy expenditure. *Int J Obes* 39: 1300–1309, 2015.

159. **Zhao P**, **Sharir H**, **Kapur A**, **Cowan A**, **Geller EB**, **Adler MW**, **Seltzman HH**, **Reggio PH**, **Heynen-Genel S**, **Sauer M**, **Chung TDY**, **Bai Y**, **Chen W**, **Caron MG**, **Barak LS**, **Abood ME**. Targeting of the orphan receptor GPR35 by pamoic acid: a potent activator of extracellular signal-regulated kinase and β-arrestin2 with antinociceptive activity. *Mol Pharmacol* 78: 560–568, 2010.

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1066 **Table 1 - Relevant kynurenines: concentrations and biological relevance**

	Enzyme/reaction	Enzyme expressed in skeletal muscle*	Receptor/biological activity	Plasma concentration (5th-95th percentile) adults (142) †
Kynurenine (KYN)	IDO/TDO	Low	AhR (90, 100)	0,94 -2,09 (μM)
Kynurenic acid (KYNA)	KAT1 KAT2 KAT3 KAT4	↑↑^ Low ↑↑^ ↑^	AhR (23) N-methyl-D-aspartate receptor (NMDAR) and α7 nicotinic acetylcholine receptor (α7nAChR) antagonist (9, 135) GPR35 agonist (153) α-amino-3-hydroxy-5-methyl-4-isoxazole propionic acid (AMPA) receptor modulator (109)	24,1 – 72,9 (nM)
3-hydroxy-L-kynurenine (3-HK)	KMO	Low	Generation of ROS (28, 137, 146)	17,4 – 50,5 (nM)
Anthranilic acid (AA)	Kynureninase	↑	Generation of ROS (146) Used as a scaffold for derivates with biological activity (89, 112, 148)	7,7 – 22,2 (nM)
3-hydroxy anthranilic acid (3-HAA)	Kynureninase	↑	Inhibit mitochondrial respiration (high doses, (111) Pro-apoptotic (type 1 T helper cells) and stimulant of proliferation (type 2 T helper cells) (34) ROS production (43, 56)	17,8 – 60,2 (nM)
Xanthurenic acid (XA)	KATs		Type-2 and type-3 metabotropic glutamate receptor (indirect interaction (35, 36)) Vesicular glutamate transport (VGLUT) (95))	7 – 33,8 (nM)
Cinnabarinic acid (CA)	dimerization of 3-hydroxyanthranilic acid (3-HAA) upon a two-step oxidation reaction (56, 78)	NP	AhR (low micromolar range) (69, 78) Type-4 metabotropic glutamate receptor (37) ROS generation (56)	ND

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1068 ^highly inducible by PGC1a1

1069 *www.proteinatlas.org

1070 † Data are from Theofylaktopoulou et al. (142)

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Figure Legends

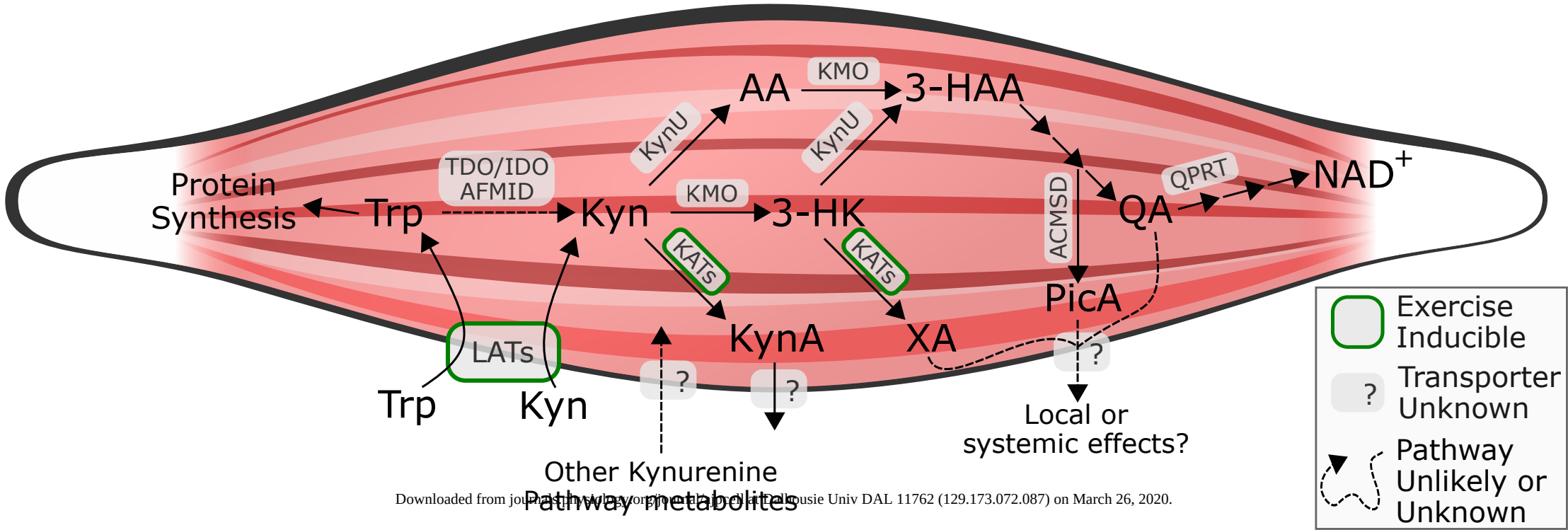
Fig. 1.

The kynurenine pathway of tryptophan degradation in skeletal muscle. Trp, Tryptophan; Kyn, Kynurenine; 3-HK, 3-Hydroxykynurenine; KynA, Kynurenic acid; AA, Anthranilic acid; 3-HAA, 3-Hydroxyanthranilic acid; PicA, Picolinic acid; QA, Quinolinic acid. TDO, Tryptophan 2,3-dioxygenase. IDO, Indoleamine 2,3-dioxygenase. AFMID, Arylformamidase. KynU, Kynureninase. KATs, Kynurenine aminotransferases. KMO, Kynurenine 3-monooxygenase. ACMSD, Aminocarboxymuconate semialdehyde decarboxylase. QPRT, Quinolinate phosphoribosyltransferase. LATs, large neutral amino acid transporters.

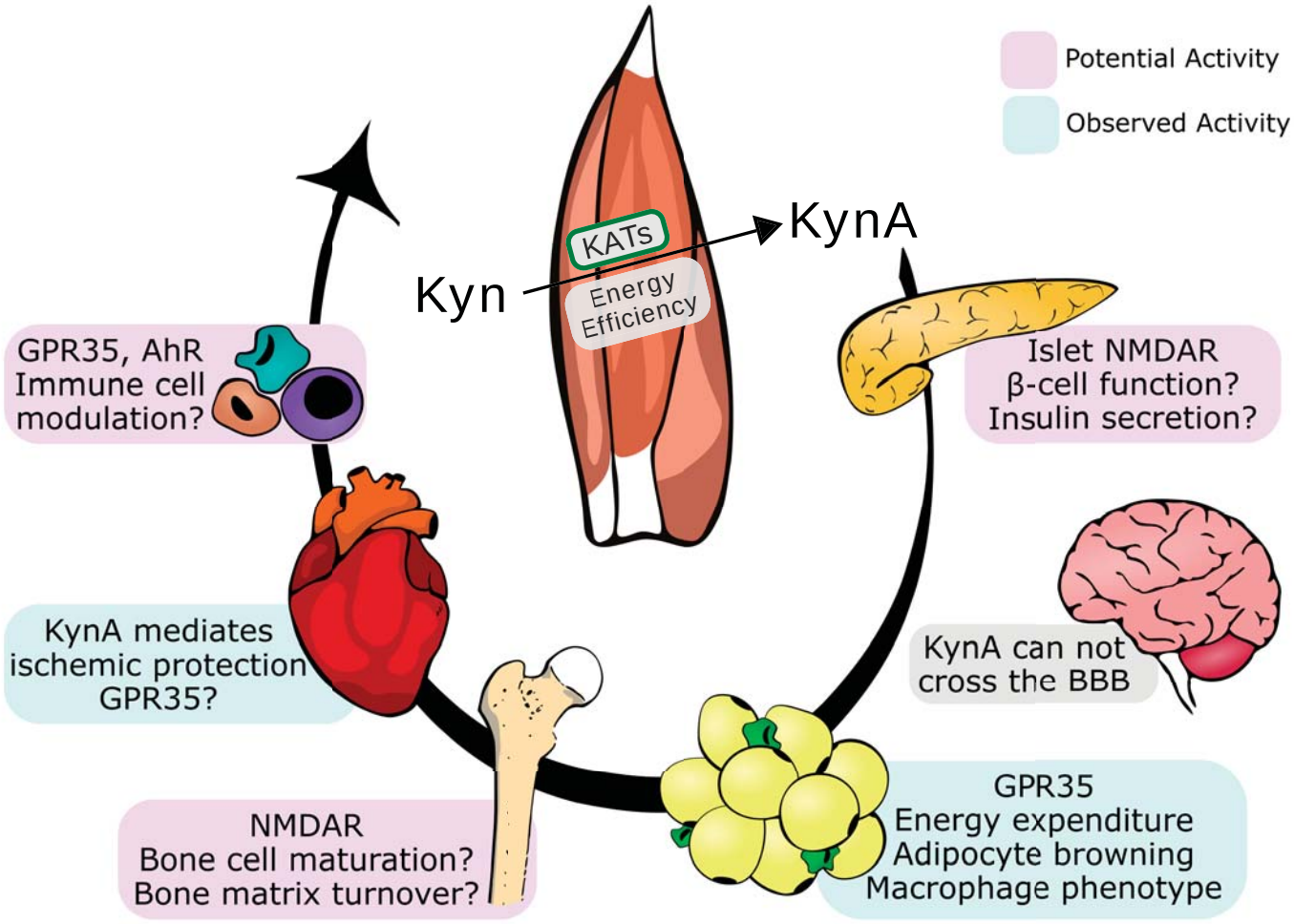
Fig. 2.

Known and potential biological activities of circulating kynurenic acid (KynA) in peripheral tissues. The conversion of kynurenine (Kyn) to KynA in exercised skeletal muscle is catalyzed by kynurenine aminotransferases (KATs) and increases energy efficiency. Exercise elevates KynA in circulation and the presence of KynA receptors in diverse tissues suggest different biological activities. (?) indicates potential activity. NMADR, N-methyl-D-aspartate receptor; BBB, Blood-brain barrier; AhR, Aryl hydrocarbon receptor.

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